

Radu Briciu

BSc, PgDip

Bucharest, Romania | EU citizen (no work permit required in the EU)

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PROFILE

Finance graduate with two years of self-directed work on stochastic-volatility calibration and derivatives pricing in Python and QuantLib, a working paper on neural-network option pricing, and a twelve-month placement in pharmaceutical business finance. Seeking a market risk, model validation, valuation or quantitative development analyst role.

EDUCATION

Postgraduate Diploma in Quantitative Finance City of London, United Kingdom

Bayes Business School (formerly Cass) – City, University of London

Sept. 2023 — Jun. 2025

- a. Dissertation: *Neural Networks for Exotic Option Pricing*, supervised by Prof. Aleš Černý; published as an SSRN working paper (see below)

Bachelor of Science in Finance, Upper Second Class Honours Westminster, United Kingdom

Westminster Business School – University of Westminster

Aug. 2019 — Jun. 2023

- a. Sandwich degree with a twelve-month placement; Mathematics for Finance 82%, Financial Modelling and Statistics 75%, Investments and Portfolio Management 75%, Dissertation 73%

EXPERIENCE

Finance & Operations Placement London, United Kingdom

Novo Nordisk UK Ltd

Aug. 2021 — Aug. 2022

- a. Twelve-month industrial placement split between the Commercial Excellence and Business Finance Analytics & Pricing teams
- b. Built sales-versus-target and cost-variance models in Excel and Power Query, and automated recurring reporting with VBA
- c. Analysed market dynamics, brand performance and customer segmentation and presented findings to both teams; reconciled financial accounts, cleaned CRM data and collected competitor prices

PROJECTS

Heston and Bates calibration to SPX option trades github.com/boomelage/heston-calibration

Daily Heston (1993) and Bates (1996) calibration to CBOE SPX intraday option trades (2012–2024) with QuantLib: a pooled, moneyness-normalised implied-vol surface per trading day, box-constrained Levenberg–Marquardt from multiple data-seeded starts, an implied-vol-space acceptance gate rejecting boundary-pegged fits, and Feller and cross-day stability diagnostics. Heston: 3,108 of 3,217 trading days accepted (96.6%), median implied-vol RMSE 0.38 vol points. Hand-built prototype (2024), hardened with AI-assisted tooling.

ML option pricing (dissertation) github.com/boomelage/machine-learning-option-pricing

Feed-forward networks trained to approximate QuantLib Heston prices of barrier and Asian options on SPX, with training data generated from about 1,600 historical Heston calibrations (2012–2024); pricing errors below 1% at up to 99.8% lower computation time. Basis of the working paper below.

Heston simulation github.com/boomelage/heston-sim

Derivation and NumPy implementation of the Broadie–Kaya (2006) exact-variance scheme with Andersen’s (2008) trapezoidal integrated-variance approximation, following Prof. Ioannis Kyriakou’s lectures: exact non-central chi-squared variance draws and a closed-form log-price update.

Realised volatility github.com/boomelage/realised-volatility

Realised variance of JPM from one-minute prices (2006–2009) at 1, 5, 10, 15-minute and daily frequencies in pandas, compared against GARCH(1,1) conditional variance estimated in OxMetrics.

PUBLICATIONS

Briciu, Radu (Sept. 2024). “Neural Networks for Exotic Option Pricing”. SSRN working paper; MSc dissertation, Bayes Business School. DOI: [10.2139/ssrn.5104328](https://doi.org/10.2139/ssrn.5104328).

SKILLS

Programming	Python 3 (NumPy, pandas, SciPy, scikit-learn, Keras, QuantLib) · MATLAB · L ^A T _E X · exposure to C and Rust
Finance and data	Bloomberg Terminal (OMON, VCUBE, historical data) · Excel · Power Query · VBA · OxMetrics · Alpha Vantage API
Tooling	Git and GitHub · Bash · PowerShell · WSL · Jupyter
Languages	Romanian (native) · English (fluent) · German (A1)

SELECTED COURSEWORK

Bayes Derivatives · Fixed Income · Numerical Methods: Applications · Econometrics of Financial Markets · Risk Analysis · Blockchain, Cryptocurrencies and Decentralized Finance

Westminster Mathematics for Finance · Financial Modelling and Statistics · Risk Management · Investments and Portfolio Management · Securities Analysis · Corporate Finance · Advanced Corporate Finance